One process, several predictive coordinate systems

Final synthesis · 12–15 minutes core · optional 8–10 minute WFA branch

ILIAD Intensive · Computational Mechanics

Start only from what the room has reached

PACING GATE

Full synthesis: use after Notebook 3’s core factorization.

If Hankel/PSR was dropped: show only the transformer-evidence slide, the caveats, and the exit.

WFA branch: open only if the factorization is secure and 8–10 minutes remain.

Which distinctions between histories change the distribution over futures?

Four lenses on predictive state

HMM belief

Posterior over a chosen model's hidden states.

Update: normalize a stochastic propagation.

Conditional Hankel row

Probabilities of all future tests.

Update: condition after appending a symbol.

Finite PSR

Probabilities of selected spanning tests.

Update: generally rational after normalization.

WFA state

Coordinates in a finite linear basis.

Update: linear symbol operators.

Different coordinates can describe the same observable predictive distinctions.

What survives a change of coordinates?

Every valid realization assigns the same probability to every finite word.

> $\Pr(w)$ is invariant; coordinates, update matrices, and arbitrary metric geometry need not be.

Invariant

word probabilities and full predictive equivalence

Representation-dependent

state labels, redundant directions, and basis entries

Requires care

Euclidean geometry after an arbitrary invertible transform

What the transformer evidence supports

1. Belief coordinates are **affinely decodable** from activations in the studied toy processes. — **supported**
2. Structure changes with training and survives held-out and shuffle comparisons. — **supported**
3. RRXOR geometry distinguishes histories beyond current optimal next-token distributions. — **supported**
4. The decoded directions are used causally to implement Bayesian updating. — **not directly tested**

Shai et al., NeurIPS 2024, arXiv:2405.15943.

Five caveats worth retaining

1 · Belief is model-relative
An HMM posterior depends on the chosen realization.

2 · One step is not every future
Matched next-token predictions can hide longer-future differences.

3 · A finite block is local evidence
Its dependencies need not determine full Hankel rank.

4 · Linear dimension is not HMM size
A finite linear realization need not be nonnegative and stochastic.

5 · Decodability is not causal use
A diagnostic map can reveal information without proving the model's forward computation relies on that coordinate.

The day’s arc

observable process → **HMM belief / MSP** → **belief geometry** → **transformer probe** → **observable predictive coordinates**

1. State one claim from today you would now defend precisely.
2. Name one experiment or observation that would change your mind.
3. Which representation made the predictive problem easiest to see—and what did it hide?

Optional branch: append a symbol linearly

OPEN ONLY IF TIME REMAINS

Return to the Notebook 3 instructor demo. Give students the observable basis blocks and solved symbol operators; do not assign matrix inversion.

Basis row

A finite vector records selected observable tests after a history.

Symbol operator

Multiplying by A_x updates that linear coordinate when symbol x is appended.

$$\Pr(x_1 \cdots x_L) = \alpha A_{x_1} \cdots A_{x_L} \omega.$$

Negative entries are not negative probabilities

What an operator entry is

A coefficient in an observable linear coordinate update.

What the induced word weights must satisfy

Nonnegativity, normalization at every length, and extension consistency:

$P(w) \geq 0,\ \sum_{|w|=L} P(w) = 1$, and $P(u) = \sum_x P(ux)$.

A weighted finite automaton is a linear realization of the process—not automatically a hidden Markov model.

Three questions worth carrying forward

1. Which predictive coordinates are actually used causally inside a trained model?
2. Which parts of representation geometry survive principled changes of basis?
3. When does finite predictive dimension admit a compact nonnegative realization?

CLOSING INVARIANT

Whatever coordinates we choose, the observable process is the set of probabilities assigned to finite words.

longum iter est per praecepta, breve et efficax per exempla. multum interest utrum non velit an nesciat. velle non discitur.

— Seneca the Younger, adapted from Moral Letters to Lucilius 6.5, 90.46, and 81.13

Primary empirical source: Shai et al., NeurIPS 2024. Technical derivations and all project-authored figures are in the three worked notebooks.